

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science & Engineering

Program: B.Sc. in CSE

Mid Term Examination, Spring 2025

Course Title: Coordinate Geometry, Linear Algebra and Vector Analysis

Course Code: MATH0541211

Marks: 20

Time: 1(one) hour

(Answer all questions)

Q.No.	Questions	Marks
1 a)	If the area of the quadrilateral formed by the points $(1, 3)$, $(2, -5)$, $(6, -2)$ and $(5, k)$ taken in order be 30, show that $k = 4$.	2
b)	Change the equation to polar coordinates $x^4 + x^2y^2 - (x + y)^2 = 0$ and show that $r = \pm(1 + \tan\theta)$	3
c)	The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x^2 + 2y^2 = 1$ When referred to rectangular axes through the point $(2, 3)$. Find the inclination of the later axes to the former.	5
2 a)	Find the equations of the bisectors of the angles between the lines represented by: $2x^2 + 7xy + 6y^2 + 13x + 22y + 20 = 0$	5
b)	Determine the nature of the equation and reduce to the standard form $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$	5